

Data Structure

Course No. 40989

Week #14

Hash Tables

Bumsoo Kim

School of Computer Science & Engineering

E-mail : bumsoo@cau.ac.kr

Office : 310-430

Hash Tables

Motivation

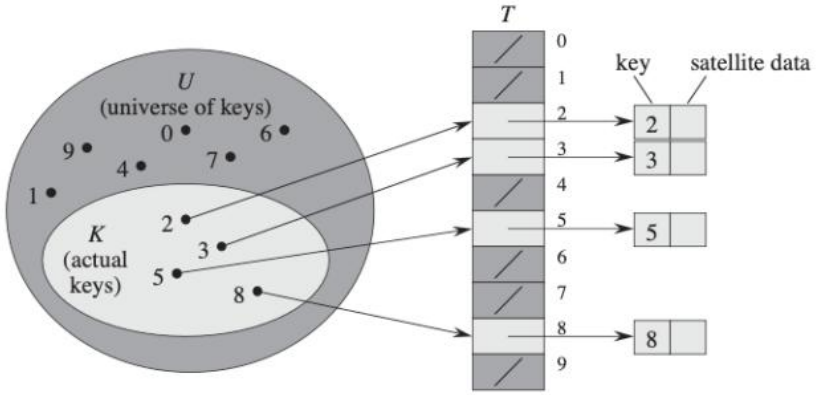
	Arrays		Linked-lists		Balanced search trees	Hash tables (avg)
	Not sorted	Sorted	Not sorted	Sorted		
Search(x)	$O(n)$	$O(\lg n)$	$O(n)$	$O(n)$	$O(\lg n)$	$\Theta(1)$
Insert(x)	$\Theta(1)$	$O(n)$	$\Theta(1)$	$O(n)$	$O(\lg n)$	$\Theta(1)$
Insert(x) (dup search)	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(\lg n)$	$\Theta(1)$
Delete(i)	$\Theta(1)$	$O(n)$	$\Theta(1)$ (DLL)	$\Theta(1)$ (DLL)	$O(\lg n)$	$\Theta(1)$
Delete(x)	$O(n)$	$O(n)$	$O(n)$	$O(n)$	$O(\lg n)$	$\Theta(1)$

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Motivation

Previous: Direct Address Tables

- Many applications require a dynamic set that supports only the dictionary operations (INSERT, SEARCH, and DELETE)
- Store key k into slot k



DIRECT-ADDRESS-SEARCH(T, k) $O(1)$
1 **return** $T[k]$

DIRECT-ADDRESS-INSERT(T, x) $O(1)$
1 $T[x.key] = x$

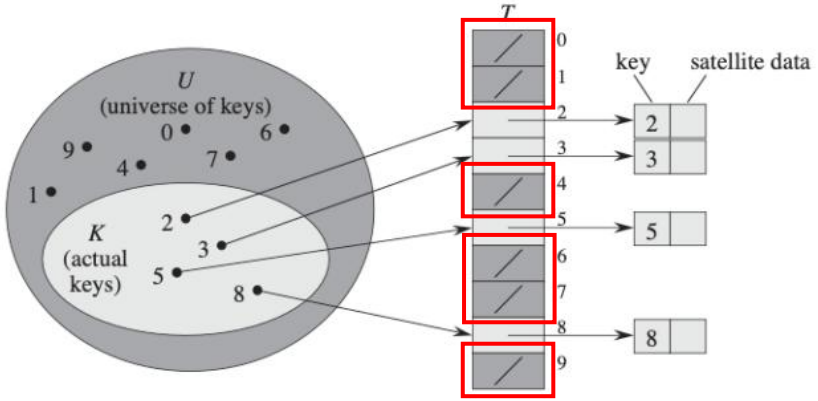
DIRECT-ADDRESS-DELETE(T, x) $O(1)$
1 $T[x.key] = \text{NIL}$

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Motivation

Previous: Direct Address Tables

- Many applications require a dynamic set that supports only the dictionary operations (INSERT, SEARCH, and DELETE)
- Store key k into slot k
- Limitations: Each element much have a distinct key
- If |K|/|U| is small, most of the slots are ***wasted***.
  - Moreover, storing a table T of size |U| may be impractical (even impossible) given the limited memory of a computer.
  - How can we reduce the space complexity to O(|K|) while preserving the time complexity O(1)?



DIRECT-ADDRESS-SEARCH(T, k) — $O(1)$
1 **return** $T[k]$

DIRECT-ADDRESS-INSERT(T, x) — $O(1)$
1 $T[x.key] = x$

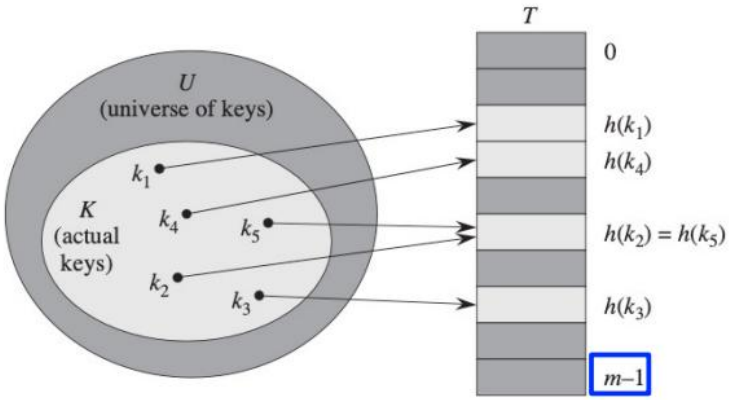
DIRECT-ADDRESS-DELETE(T, x) — $O(1)$
1 $T[x.key] = \text{NIL}$

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Motivation

What is hashing?

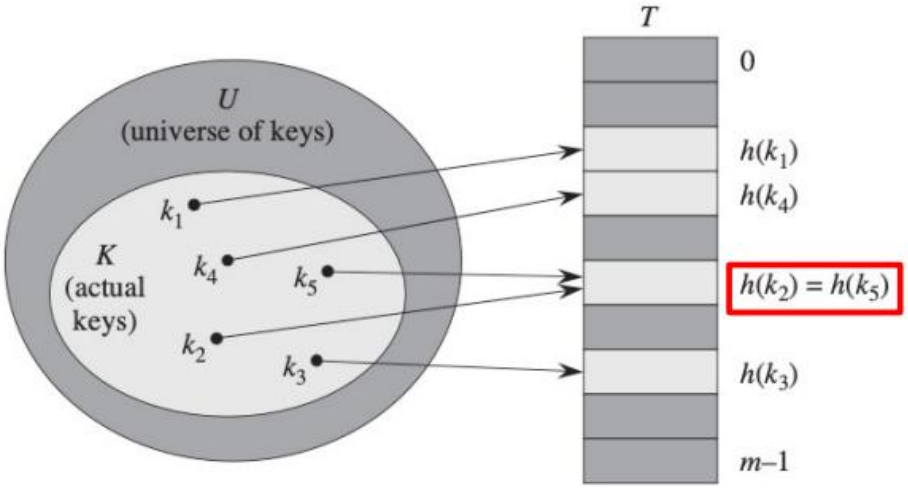
- Storing key k in slot $h(k)$, not slot k, by using a **hash function** h
- The hash function h maps the universe of keys U into the slots of a hash table T
- Therefore, the space requirement become identical to the size of a hash table T



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Hash Function

- However, hash collision necessarily occurs
  - Two keys k_1 and k_2 where $k_1 \neq k_2$ but $h(k_1) = h(k_2)$ under **hash function** h
  - Two keys k_1 and k_2 where $k_1 \neq k_2$, $h(k_1) \neq h(k_2)$, but $h(k_1)\ \%\ m = h(k_2)\ \%\ m$
  - Avoiding collisions is impossible when $|U| > m$



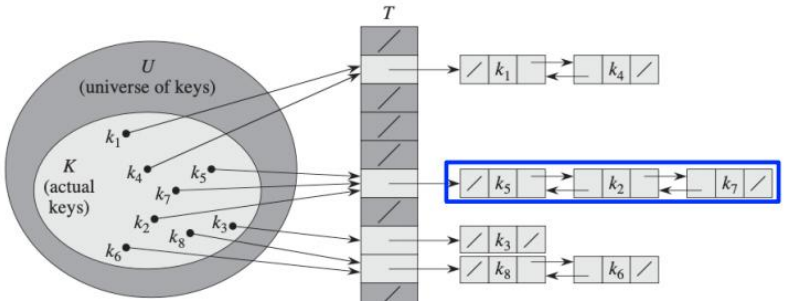
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Hash Function

Collision resolution by Chaining

- Maintaining a linked list for each slot to store keys

- Insertion: $O(1)$
- Deletion: $O(1)$
  - It takes an element x as input, not its key k (we do not need to search for x)
- Search: $O(l)$
  - Proportional to the length of the list l (depending on a hash function)



CHAINED-HASH-INSERT(T, x)
1 insert x at the head of list $T[h(x.key)]$

CHAINED-HASH-SEARCH(T, k)
1 search for an element with key k in list $T[h(k)]$

CHAINED-HASH-DELETE(T, x)
1 delete x from the list $T[h(x.key)]$

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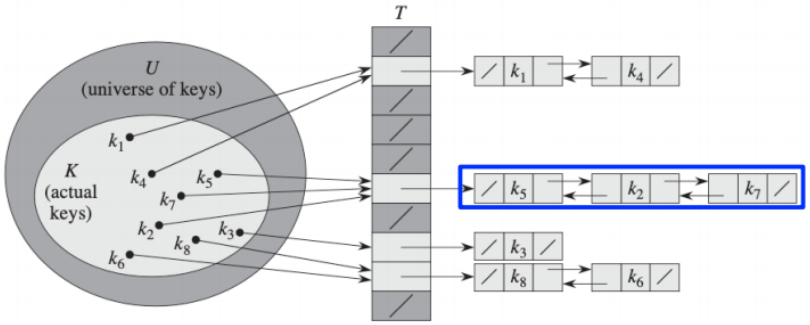
Hash Function

Collision resolution by Chaining

- Maintaining a linked list for each slot to store keys

- Worst-case complexity: $O(n)$ → when?
- Average-case complexity: $O(1+\alpha)$ (computing the hash function + searching)
  - Load factor: $\alpha = n/m$ (average length of a chain)
    - n is the number of elements in a hash table T and m is the number of slots in the table

What does this analysis mean? → Search operation can run in $O(1)$ time



CHAINED-HASH-INSERT(T, x)
1 insert x at the head of list $T[h(x.key)]$

CHAINED-HASH-SEARCH(T, k)
1 search for an element with key k in list $T[h(k)]$

CHAINED-HASH-DELETE(T, x)
1 delete x from the list $T[h(x.key)]$

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Hash Function

Example

Let's say that we are making a hash table for the phone numbers of SKT T1 players.

- 020-1111-1234 Faker
- 020-1211-5611 Oner
- 020-3011-1341 Gumayushi
- 020-9910-4123 Keria
- 020-8000-1262 Doran

Lets take our hash function as a 'sum of all the digits', and we have a hash table with a size 3.

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Hash Function

Example

Let's say that we are making a hash table for the phone numbers of SKT T1 players.

- 020-1111-1234 Faker
- 020-1211-5611 Oner
- 020-3011-1341 Gumayushi
- 020-9910-4123 Keria
- 020-8000-1262 Doran

Lets take our hash function as a 'sum of all the digits', and we have a hash table with a size 3.

- 020-1111-1234 Faker = 16, hash index 16 mod 3 = 1
- 020-1211-5611 Oner = 20, hash index 20 mod 3 = 2
- 020-3011-1341 Gumayushi = 16, hash index 16 mod 3 = 1
- 020-9910-4123 Keria = 31, hash index 31 mod 3 = 1
- 020-8000-1262 Doran = 21, hash index 21 mod 3 = 0

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Hash Function

Caution

- When building a hash table, avoid m being a power of 2 (e.g., 2^p)
- If so, then is $h(k)$ is just the p lowest-order bits of k.

- In other word, the hash function depends on only lower-order p-bits, instead of all the bits

$k = 10110100, m = 2^4 = 00010000$

$h(k) = k \bmod 2^4 = 0100$

$k = 28237481, m = 10^4 = 10000$

$h(k) = k \bmod 10^4 = 7481$

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Hash Function

Collision resolution by Open Addressing

- Storing all elements in the hash table itself
- Each slot contains either a key or NIL
- Advantages?
  - Do not need pointers
  - Extra memory provides the hash table with larger number of slots. Fewer collisions, faster retrieval.

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Hash Function

Collision resolution by Open Addressing

- Storing all elements in the hash table itself
- Each slot contains either a key or NIL
- Advantages?
  - Do not need pointers
  - Extra memory provides the hash table with larger number of slots. Fewer collisions, faster retrieval.

- Examine the hash table (***probe***) until it finds an empty slot (insert) or the key (search)
- The probe sequence for every key k
  - $h(k,0), h(k,1), \ldots, h(k,m-1)$

```
HASH-INSERT(T, k)
1  i = 0
2  repeat
3      j = h(k, i)
4      if T[j] == NIL
5          T[j] = k
6          return j
7      else i = i + 1
8  until i == m
9  error "hash table overflow"
```

```
HASH-SEARCH(T, k)
1  i = 0
2  repeat
3      j = h(k, i)
4      if T[j] == k
5          return j
6      i = i + 1
7  until T[j] == NIL or i == m
8  return NIL
```

$m = 13$,
$k = \{5,14,29,25,17,21,18,32,20,9,15,27\}$
$h(k) = k \bmod 13$

0		
1	14	27
2	15	
3	29	
4	17	
5	5	18
6		32
7		20
8	21	
9		9
10		
11		
12	25	

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Hash Function

Collision resolution by Open Addressing

- Storing all elements in the hash table itself
- Each slot contains either a key or NIL
- Advantages?
  - Do not need pointers
  - Extra memory provides the hash table with larger number of slots. Fewer collisions, faster retrieval.

- Can we remove the key just by putting NIL into the slot?
  - After inserting key k into slot j, we inserted key k' into another slot
    - During the insertion, we had probed slot j
  - Delete key k by storing NIL into slot j
  - When we want to search for key k', we would probe slot j before probing the slot where key k' was eventually stored
  - Thus, we fail to search for key k' even though k' is in the table
- Solution: Mark the slot by the special value "DELETED"
  - Search treats DELETED as though the slot holds a key that does not match the one being searched for
  - Insertion treats DELETED as though the slot is empty to reuse it

$m = 13$,
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$h(k) = k \bmod 13$

0		
1	14	27
2	15	
3	29	
4	17	
5	5	18
6		32
7		20
8	21	
9		9
10		
11		
12	25	

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Hash Function

Collision resolution by Open Addressing

- Three methods : Linear Probing / Quadratic Probing / Double Hashing

$m = 13,$
$k = \{5,14,29,25,17,21,18,32,20,9,15,27\}$
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5	5	18
6		32
7		20
8	21	
9		9
10		
11		
12	25	

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Hash Function

Collision resolution by Open Addressing

- Three methods : **Linear Probing** / Quadratic Probing / Double Hashing

$$h' : U \rightarrow \{0,1, \ldots, m-1\}$$

$$h(k,i) = (h'(k) + i) \bmod m$$

```
HASH-INSERT(T, k)
1  i = 0
2  repeat
3      j = h(k, i)
4      if T[j] == NIL
5          T[j] = k
6          return j
7      else i = i + 1
8  until i == m
9  error "hash table overflow"
```

```
HASH-SEARCH(T, k)
1  i = 0
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```

$m = 13$,
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0		
1	14	27
2	15	
3	29	
4	17	
5	5	18
6		32
7		20
8	21	
9		9
10		
11		
12	25	

$m = 13$,
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$h(k) = k \bmod 13$

$h(k,i) = (k + i) \bmod 13$

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Hash Function

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```
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7  until T[j] == NIL or i == m
8  return NIL
```

This method suffers from a problem known as ***primary clustering:***
Long runs of occupied slots build up, increasing the average search time

$m = 13,$
$k = \{5,14,29,25,17,21,18,32,20,9,15,27\}$
$h(k) = k \bmod 13$

0		
1	14	27
2	15	
3	29	
4	17	
5	5	18
6		32
7		20
8	21	
9		9
10		
11		
12	25	

$m = 13,$
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Hash Function

Collision resolution by Open Addressing

- Three methods : Linear Probing / **Quadratic Probing** / Double Hashing

$$h' : U \rightarrow \{0,1,\ldots,m-1\}$$

$$h(k,i) = (h'(k) + c_1 i + c_2 i^2) \bmod m$$

HASH-INSERT(T,k)

```
1  i = 0
2  repeat
3      j = h(k,i)
4      if T[j] == NIL
5          T[j] = k
6          return j
7      else i = i + 1
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HASH-SEARCH(T,k)

```
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2	15	
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4	17	
5	5	18
6	32	
7	20	
8	21	
9		9
10		
11		
12	25	

$m = 13$,
$k = \{5,14,29,25,17,21,18,32,20,9,15,27\}$
$h(k) = k \bmod 13$

$h(k,i) = (k + i + 3i^2) \bmod 13$

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Hash Function

Collision resolution by Open Addressing

- Three methods : Linear Probing / **Quadratic Probing** / Double Hashing

$$h' : U \rightarrow \{0,1,\ldots,m-1\}$$

$$h(k,i) = (h'(k) + c_1 i + c_2 i^2) \bmod m$$

HASH-INSERT(T,k)

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HASH-SEARCH(T,k)

```
1  i = 0
2  repeat
3      j = h(k,i)
4      if T[j] == k
5          return j
6      i = i + 1
7  until T[j] == NIL or i == m
8  return NIL
```

Jump to avoid primary clustering

However, it still have a problem known as ***secondary clustering:***
If two keys have the same h' value, then they have the same probe sequence

$m = 13$,
$k = \{5,14,29,25,17,21,18,32,20,9,15,27\}$
$h(k) = k \bmod 13$

0		
1	14	27
2	15	
3	29	
4	17	
5	5	18
6	32	
7	20	
8	21	
9		9
10		
11		
12	25	

$m = 13$,
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$h(k) = k \bmod 13$

$h(k,i) = (k + i + 3i^2) \bmod 13$

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Hash Function

Collision resolution by Open Addressing

- Three methods : Linear Probing / Quadratic Probing / **Double Hashing**

$$h(k,i) = (h_1(k) + i \cdot h_2(k)) \bmod m$$

```
HASH-INSERT(T, k)
1  i = 0
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3      j = h(k, i)
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10		
11		
12	25	

$m = 13$,
$k = \{5,14,29,25,17,21,18,32,20,9,15,27\}$
$h_1(k) = k \bmod 13$
$h_2(k) = 1 + (k \bmod 11)$

$$h(k,i) = (h_1(k) + i \cdot h_2(k)) \bmod 13$$

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Hash Function

Open Addressing Exercise

- Consider inserting the keys 10, 22, 31, 4, 15, 28, 17, 88, 59 into a hash table of length m=11 using open addressing with the auxiliary hash function $h'(k) = k$. Illustrate the result of inserting these keys using (1) linear probing, (2) using quadratic probing with $c_1 = 1$ and $c_2 = 3$, and (3) double hashing with $h_1(k) = k$ and $h_2(k) = 1 + (k \bmod (m-1))$

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Hash Function

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10 mod 11 = 10

0 mod 11	
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	
10 mod 11	10

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Hash Function

Open Addressing Exercise

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22 mod 11 = 0

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	
10 mod 11	10

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Hash Function

Open Addressing Exercise

- Consider inserting the keys 10, 22, 31, 4, 15, 28, 17, 88, 59 into a hash table of length m=11 using open addressing with the auxiliary hash function $h'(k) = k$. Illustrate the result of inserting these keys using (1) linear probing, (2) using quadratic probing with $c_1 = 1$ and $c_2 = 3$, and (3) double hashing with $h_1(k) = k$ and $h_2(k) = 1 + (k \bmod (m-1))$

31 mod 11 = 9

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	31
10 mod 11	10

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Hash Function

Open Addressing Exercise

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4 mod 11 = 4

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	31
10 mod 11	10

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Hash Function

Open Addressing Exercise

- Consider inserting the keys 10, 22, 31, 4, 15, 28, 17, 88, 59 into a hash table of length m=11 using open addressing with the auxiliary hash function $h'(k) = k$. Illustrate the result of inserting these keys using (1) linear probing, (2) using quadratic probing with $c_1 = 1$ and $c_2 = 3$, and (3) double hashing with $h_1(k) = k$ and $h_2(k) = 1 + (k \bmod (m-1))$

15 mod 11 = 4

0 mod 11	22	
1 mod 11		
2 mod 11		
3 mod 11		
4 mod 11	4	15
5 mod 11		
6 mod 11		
7 mod 11		
8 mod 11		
9 mod 11	31	
10 mod 11	10	

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Hash Function

Open Addressing Exercise

- Consider inserting the keys 10, 22, 31, 4, 15, 28, 17, 88, 59 into a hash table of length m=11 using open addressing with the auxiliary hash function $h'(k) = k$. Illustrate the result of inserting these keys using (1) linear probing, (2) using quadratic probing with $c_1 = 1$ and $c_2 = 3$, and (3) double hashing with $h_1(k) = k$ and $h_2(k) = 1 + (k \bmod (m-1))$

15 mod 11 = 4

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	15
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	31
10 mod 11	10

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Hash Function

Open Addressing Exercise

- Consider inserting the keys 10, 22, 31, 4, 15, 28, 17, 88, 59 into a hash table of length m=11 using open addressing with the auxiliary hash function $h'(k) = k$. Illustrate the result of inserting these keys using (1) linear probing, (2) using quadratic probing with $c_1 = 1$ and $c_2 = 3$, and (3) double hashing with $h_1(k) = k$ and $h_2(k) = 1 + (k \bmod (m-1))$

28 mod 11 = 6

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	15
6 mod 11	28
7 mod 11	
8 mod 11	
9 mod 11	31
10 mod 11	10

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Hash Function

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17 mod 11 = 6

0 mod 11	22	
1 mod 11		
2 mod 11		
3 mod 11		
4 mod 11	4	
5 mod 11	15	
6 mod 11	28	17
7 mod 11		
8 mod 11		
9 mod 11	31	
10 mod 11	10	

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Hash Function

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17 mod 11 = 6

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	15
6 mod 11	28
7 mod 11	17
8 mod 11	
9 mod 11	31
10 mod 11	10

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88 mod 11 = 0

0 mod 11	22 88
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	15
6 mod 11	28
7 mod 11	17
8 mod 11	
9 mod 11	31
10 mod 11	10

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88 mod 11 = 0

0 mod 11	22
1 mod 11	88
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	15
6 mod 11	28
7 mod 11	17
8 mod 11	
9 mod 11	31
10 mod 11	10

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59 mod 11 = 4

0 mod 11	22	
1 mod 11	88	
2 mod 11		
3 mod 11		
4 mod 11	4	59
5 mod 11	15	
6 mod 11	28	
7 mod 11	17	
8 mod 11		
9 mod 11	31	
10 mod 11	10	

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59 mod 11 = 4

0 mod 11	22	
1 mod 11	88	
2 mod 11		
3 mod 11		
4 mod 11	4	
5 mod 11	15	59
6 mod 11	28	
7 mod 11	17	
8 mod 11		
9 mod 11	31	
10 mod 11	10	

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59 mod 11 = 4

0 mod 11	22	
1 mod 11	88	
2 mod 11		
3 mod 11		
4 mod 11	4	
5 mod 11	15	
6 mod 11	28	59
7 mod 11	17	
8 mod 11		
9 mod 11	31	
10 mod 11	10	

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59 mod 11 = 4

0 mod 11	22	
1 mod 11	88	
2 mod 11		
3 mod 11		
4 mod 11	4	
5 mod 11	15	
6 mod 11	28	
7 mod 11	17	59
8 mod 11		
9 mod 11	31	
10 mod 11	10	

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59 mod 11 = 4

0 mod 11	22
1 mod 11	88
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	15
6 mod 11	28
7 mod 11	17
8 mod 11	59
9 mod 11	31
10 mod 11	10

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10 mod 11 = 10

0 mod 11	
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	
10 mod 11	10

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Hash Function

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22 mod 11 = 0

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	
10 mod 11	10

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31 mod 11 = 9

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	31
10 mod 11	10

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4 mod 11 = 4

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	31
10 mod 11	10

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15 mod 11 = 4

0 mod 11	22	
1 mod 11		
2 mod 11		
3 mod 11		
4 mod 11	4	15
5 mod 11		
6 mod 11		
7 mod 11		
8 mod 11		
9 mod 11	31	
10 mod 11	10	

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Hash Function

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15 mod 11 = 4, probe with $i = 1$, $h(15,1) = \left(4 + 1 + 3(1^2)\right) \bmod 11 = 8$

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	15
9 mod 11	31
10 mod 11	10

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28 mod 11 = 6

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	
6 mod 11	28
7 mod 11	
8 mod 11	15
9 mod 11	31
10 mod 11	10

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17 mod 11 = 6

0 mod 11	22	
1 mod 11		
2 mod 11		
3 mod 11		
4 mod 11	4	
5 mod 11		
6 mod 11	28	17
7 mod 11		
8 mod 11	15	
9 mod 11	31	
10 mod 11	10	

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17 mod 11 = 6, probe with $i = 1$, $h(17,1) = \left(6 + 1 + 3(1^2)\right) \bmod 11 = 10$

0 mod 11	22	
1 mod 11		
2 mod 11		
3 mod 11		
4 mod 11	4	
5 mod 11		
6 mod 11	28	
7 mod 11		
8 mod 11	15	
9 mod 11	31	
10 mod 11	10	17

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17 mod 11 = 6, probe with $i = 2$, $h(17,2) = \left(6 + 2 + 3(2^2)\right) \bmod 11 = 9$

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	
6 mod 11	28
7 mod 11	
8 mod 11	15
9 mod 11	31 17
10 mod 11	10

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17 mod 11 = 6, probe with $i = 3$, $h(17,3) = \left(6 + 3 + 3(3^2)\right) \bmod 11 = 3$

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	17
4 mod 11	4
5 mod 11	
6 mod 11	28
7 mod 11	
8 mod 11	15
9 mod 11	31
10 mod 11	10

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Hash Function

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88 mod 11 = 0

0 mod 11	22 88
1 mod 11	
2 mod 11	
3 mod 11	17
4 mod 11	4
5 mod 11	
6 mod 11	28
7 mod 11	
8 mod 11	15
9 mod 11	31
10 mod 11	10

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88 mod 11 = 0, probe with $i = 1$, $h(88,1) = \left(0 + 1 + 3(1^2)\right) \bmod 11 = 4$

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	17
4 mod 11	4 88
5 mod 11	
6 mod 11	28
7 mod 11	
8 mod 11	15
9 mod 11	31
10 mod 11	10

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88 mod 11 = 0, probe with $i = 2$, $h(88,2) = \left(0 + 2 + 3(2^2)\right) \bmod 11 = 3$

0 mod 11	22	
1 mod 11		
2 mod 11		
3 mod 11	17	88
4 mod 11	4	
5 mod 11		
6 mod 11	28	
7 mod 11		
8 mod 11	15	
9 mod 11	31	
10 mod 11	10	

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Hash Function

Open Addressing Exercise

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88 mod 11 = 0, probe with $i = 3$, $h(88,3) = \left(0 + 3 + 3(3^2)\right) \bmod 11 = 8$

0 mod 11	22	
1 mod 11		
2 mod 11		
3 mod 11	17	
4 mod 11	4	
5 mod 11		
6 mod 11	28	
7 mod 11		
8 mod 11	15	88
9 mod 11	31	
10 mod 11	10	

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Hash Function

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88 mod 11 = 0, probe with $i = 4$, $h(88,4) = \left(0 + 4 + 3(4^2)\right) \bmod 11 = 4$

0 mod 11	22	
1 mod 11		
2 mod 11		
3 mod 11	17	
4 mod 11	4	
5 mod 11		
6 mod 11	28	
7 mod 11		
8 mod 11	15	88
9 mod 11	31	
10 mod 11	10	

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Hash Function

Open Addressing Exercise

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88 mod 11 = 0, probe with $i = 5$, $h(88,5) = \left(0 + 5 + 3(5^2)\right) \bmod 11 = 9$

0 mod 11	22	
1 mod 11		
2 mod 11		
3 mod 11	17	
4 mod 11	4	
5 mod 11		
6 mod 11	28	
7 mod 11		
8 mod 11	15	
9 mod 11	31	88
10 mod 11	10	

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Hash Function

Open Addressing Exercise

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88 mod 11 = 0, probe with $i = 6$, $h(88,6) = \left(0 + 6 + 3(6^2)\right) \bmod 11 = 4$

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	17
4 mod 11	4 88
5 mod 11	
6 mod 11	28
7 mod 11	
8 mod 11	15
9 mod 11	31
10 mod 11	10

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Hash Function

Open Addressing Exercise

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10 mod 11 = 10

0 mod 11	
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	
10 mod 11	10

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22 mod 11 = 0

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	
10 mod 11	10

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Hash Function

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31 mod 11 = 9

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	31
10 mod 11	10

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4 mod 11 = 4

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	31
10 mod 11	10

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15 mod 11 = 4

0 mod 11	22	
1 mod 11		
2 mod 11		
3 mod 11		
4 mod 11	4	15
5 mod 11		
6 mod 11		
7 mod 11		
8 mod 11		
9 mod 11	31	
10 mod 11	10	

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Hash Function

Open Addressing Exercise

- Consider inserting the keys 10, 22, 31, 4, 15, 28, 17, 88, 59 into a hash table of length m=11 using open addressing with the auxiliary hash function $h'(k) = k$. Illustrate the result of inserting these keys using (1) linear probing, (2) using quadratic probing with $c_1 = 1$ and $c_2 = 3$, and (3) double hashing with $h_1(k) = k$ and $h_2(k) = 1 + (k \bmod (m-1))$

15 mod 11 = 4, probe with $i = 1$, $h(15,1) = (4 + 1 \cdot (1 + 15 \bmod 10)) \bmod 11 = 10 \bmod 11 = 10$

0	mod 11	22
1	mod 11	
2	mod 11	
3	mod 11	
4	mod 11	4
5	mod 11	
6	mod 11	
7	mod 11	
8	mod 11	
9	mod 11	31
10	mod 11	10 15

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15 mod 11 = 4, probe with $i = 2$, $h(15,2) = (4 + 2 \cdot (1 + 15 \bmod 10)) \bmod 11 = 16 \bmod 11 = 5$

0 mod 11	22
1 mod 11	
2 mod 11	
3 mod 11	
4 mod 11	4
5 mod 11	15
6 mod 11	
7 mod 11	
8 mod 11	
9 mod 11	31
10 mod 11	10

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Thank you

Bumsoo Kim

School of Computer Science & Engineering

E-mail : bumsoo@cau.ac.kr

Office : 310-430